

AC6381A Datasheet

Zhuhai Jieli Technology Co.,LTD

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AC6381A Features

High performance 32-bit RISC CPU

- 32-bit DSP supports hardware Float Point Unit(FPU)
- Up to 160MHz programmable processor
- 64 Vectored interrupts
- 8 Levels interrupt priority

Flexible I/O

- 24 GPIO pins
- All GPIO pins can be programmable as input or output individually
- All GPIO pins are internal pull-up/pull-down selectable individually
- CMOS/TTL level schmitt triggered input
- External wake up/interrupt on all GPIOs

Peripheral Feature

- One Full Speed USB OTG controller
- Six Multi-function 32-bit timers, support capture and PWM mode
- Three full-duplex basic UART, support DMA mode
- Two SPI interface supports host and device mode
- One hardware IIC interface supports host and device mode
- Two Built-in low power Cap Sense Keys
- Built-in Cap Sense Key controller
- 10-bit ADC for analog sampling
- Power-on reset

Bluetooth Feature

- Dual-mode BT6.0 (DN Q334307)
- Meet class1 class2 and class3 transmitting power requirement
- Support GFSK and $\pi/4$ DQPSK all packet types
- Provides maximum +8dbm@BDR, +6dbm@EDR transmitting power
- Receiver with -94dBm@EDR sensitivity

- Fast AGC for enhanced dynamic range
- Supports a2dp\avctp\avdtp\avrcp\hfp\spp\smpt\att\gap gatt\rfcomm\sdp\l2cap profile
- a2dp 1.3\avctp 1.4\avdtp 1.3\avrcp 1.5\hfp 1.5\spp 1.0\rfcomm 1.2\pnp 1.3\hid 1.0\sdp core4.2\l2cap core 4.2

Power Supply

- Low voltage LDO for internal digital and analog circuit supply
- 2uA current consumption in the soft-off mode
- Built-in LDO for the core, I/O, Bluetooth and flash
- VBAT is 1.8V to 5.5V
- VDDIO is 1.8V to 3.4V

Packages

- QFN32(4x4mm)

Temperature

- Operating temperature
TC=-20°C to +85°C (standard range)
TC=-40°C to +105°C (extended range)
- Storage temperature: -65°C to +150°C

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1. Pin Definition

1.1 Pin Assignment

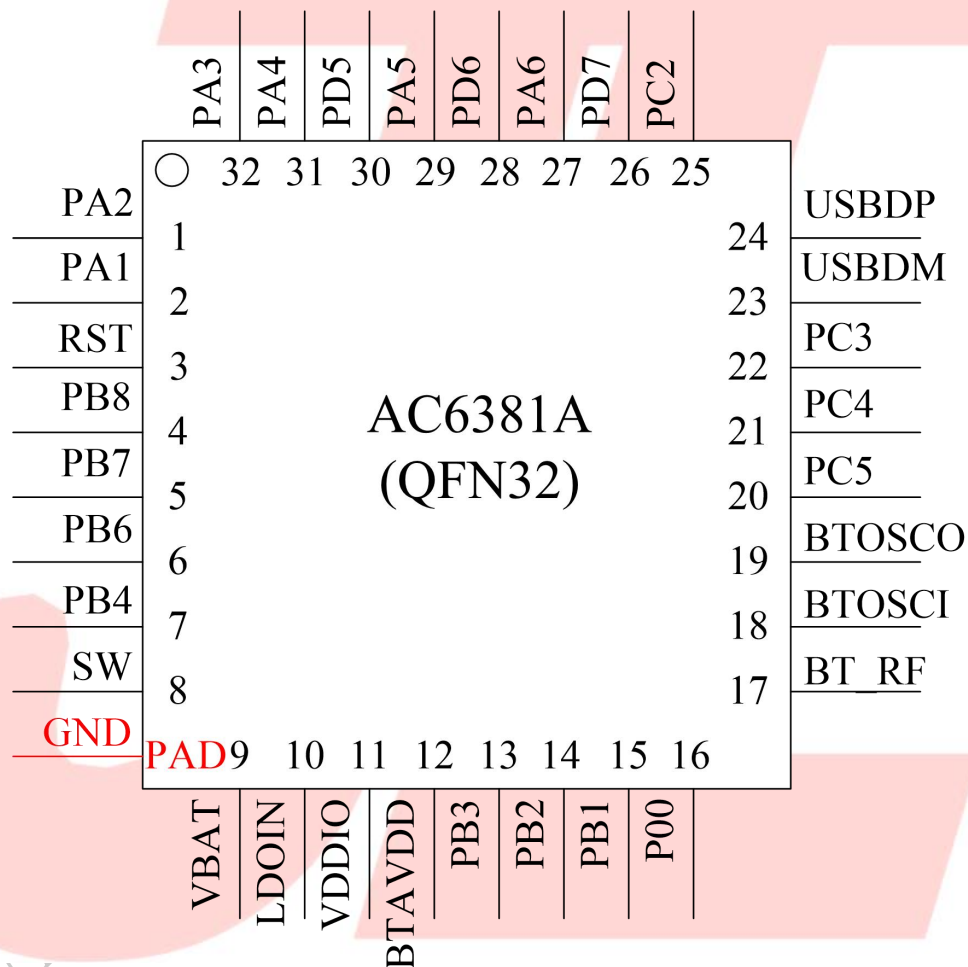


Figure 1-1 AC6381A_QFN32 Package Diagram

1.2 Pin Description

Table 1-1 AC6381A_QFN32 Pin Description

PIN NO.	Name	I/O Type	Function	Other Function
1	PA2	I/O	GPIO	CAP3: Timer3 Capture; UART1_RXC: Uart1 Data In(C);
2	PA1	I/O	GPIO	SPI1_DIC: SPI1 Data In(C); PWM0: Timer0 PWM Output; UART1_TXC: Uart1 Data Out(C);
3	RST	I	GPIO	RESET:High Active;
4	PB8	I/O	GPIO (High Voltage)	UART0_RXB: Uart0 Data In(B); CAP4: Timer4 Capture;
5	PB7	I/O	GPIO	SPI1_DOA: SPI1 Data Out(A); Q-decoder1; TOUCH1:Touch Input Channel 6 ADC8: ADC Channel 8; UART0_TXB: Uart0 Data Out(B);
6	PB6	I/O	GPIO	SPI1_CLKA: SPI1 Clock(A) ; Q-decoder0; ADC9: ADC Channel 9; TOUCH7:Touch Input Channel 7; UART1_RXA: Uart1 Data In(A); PWM2: Timer2 PWM Output;
7	PB4	I/O	GPIO	CLKOUT0; LVD:Low Voltage Detect; SPI1_DIA: SPI1 Data In(A); ADC12: ADC Channel 12; TOUCH6:Touch Input Channel 6; UAR1_TXA: Uart1 Data Out(A); TMR2: Timer2 Clock In;
8	SW	P	DC-DC Switch Pin	-
9	VBAT	P	LDO Power	-
10	LDOIN	P	Charge Power 5V	PWM3: Timer3 PWM Output; UART0_TXC: Uart0 Data Out(C); UART0_RXC: Uart0 Data In(C);
11	VDDIO	P	IO Power 3.3V	-

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12	BTA VDD	P	Core Power 1.3V	-
13	PB3	I/O	GPIO	SPI2_DIC: SPI2 Data In(C); UART1_TXB: Uart1 Data Out(B); UART1_RXB: Uart1 Data In(B); TMR4: Timer4 Clock In;
14	PB2	I/O	GPIO	SPI2_DOC: SPI2 Data Out(C); ADC7: ADC Channel 7; UART2_RXC: Uart2 Data In(C); CAP5: Timer5 Capture; LP_TH1: Low Power Touch Channel 1;
15	PB1	I/O	GPIO (pull up)	Long Press Reset; UART2_TXC: Uart2 Data Out(C); ADC6: ADC Channel 6; LP_TH0: Low Power Touch Channel 0;
16	P00	I/O	GPIO (High Voltage)	
17	BT_RF	-	RF Antenna	-
18	BTOSCI	I	BTOSCI	-
19	BTOSCO	O	BTOSCO	-
20	PC5	I/O	GPIO	SPI1_DOB: SPI1 Data Out(B); IIC_SDA_B: IIC SDA(B); ADC5: ADC Channel 5; UART2_RXD: Uart2 Data In(D);
21	PC4	I/O	GPIO	SPI1_CLKB: SPI1 Clock(B) ; IIC_SCL_B: IIC SCL(B); ADC4: ADC Channel 4; UART2_TXD: Uart2 Data Out(D); PWM4: Timer4 PWM Output;
22	PC3	I/O	GPIO	SPI1_DIB: SPI1 Data In(B); IIC_SDA_C: IIC SDA(C); ADC3: ADC Channel 3; TOUCH5: Touch Input Channel 5; UART0_RXD: Uart0 Data In(D); TMR3: Timer3 Clock In;
23	USBDM	I/O	GPIO (pull down)	SPI2_DOB: SPI2 Data Out(B); IIC_SDA_A: IIC SDA(A); ADC11: ADC Channel 11; UART1_RXD: Uart1 Data In(D);
24	USBDP	I/O	GPIO (pull down)	SPI2_CLKB: SPI2 Clock(B); IIC_SCL_A: IIC SCL(A); ADC10: ADC Channel 10; UART1_TXD: Uart1 Data Out(D);

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25	PC2	I/O	GPIO	SPI2_DIB: SPI2 Data In(B); IIC_SCL_C: IIC SCL(C); TOUCH4:Touch Input Channel 4; UART0_TXD: Uart0 Data Out(D); TMR1: Timer1 Clock In;
26	PD7	I/O	GPIO	UART2_RXB: Uart2 Data In(B); TMR5: Timer5 Clock In;
27	PA6	I/O	GPIO	UART1_RTS; SPI2_DOA: SPI2 Data Out(A); IIC_SDA_D: IIC SDA(D); ADC2: ADC Channel 2; TOUCH3:Touch Input Channel 3; UART0_RXA: Uart0 Data In(A); CAP0: Timer0 Capture;
28	PD6	I/O	GPIO	UART2_TXB: Uart2 Data Out(B); TMR0: Timer0 Clock In;
29	PA5	I/O	GPIO	UART1_CTS; SPI2_CLKA: SPI2 Clock(A); IIC_SCL_D: IIC SCL(D); ADC1: ADC Channel 1; TOUCH2:Touch Input Channel 2; UART0_TXA: Uart0 Data Out(A); PWM5: Timer5 PWM Output;
30	PD5	I/O	GPIO (Output 0)	
31	PA4	I/O	GPIO (High Voltage)	SPI2_DIA: SPI2 Data In(A); UART2_RXA: Uart2 Data In(A); CAP2: Timer2 Capture;
32	PA3	I/O	GPIO	SPI1_DOC: SPI1 Data Out(C); ADC0: ADC Channel 0; TOUCH0:Touch Input Channel 0; UART2_TXA: Uart2 Data Out(A); PWM1: Timer1 PWM Output;
PAD		P	GND	-

Note : Weak anti-static ability of PB3, need to take ESD protection measures when using.

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2. Electrical Characteristics

2.1 Absolute Maximum Ratings

Table 2-1

Symbol	Parameter	Min	Max	Unit
T _{opt}	Operating temperature	-40	+85	°C
T _{stg}	Storage temperature	-65	+150	°C
V _{BAT}	Supply Voltage	-0.3	5.5	V
LDOIN	Charge Input Voltage	-0.3	6	V
VDDIO	3.3V IO Input Voltage	-0.3	3.6	V

Note : The chip can be damaged by any stress in excess of the absolute maximum ratings listed below

2.2 Recommended Operating Conditions

Table 2-2

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
V _{BAT}	Voltage Input	1.8	3.7	5.5	V	—
LDOIN	Voltage Input	4.5	5.0	5.5	V	—
Normal mode						
VDDIO	Voltage output	-	3.0	-	V	V _{BAT} = 4.2V, 10mA loading
	Loading current	-	-	100	mA	VDDIO=3V@V _{BAT} = 4.2V
BTA _{VDD}	Voltage output	-	1.25	-	V	VDDIO=3V,10mA loading
	Loading current	-	-	60	mA	BTA _{VDD} =1.25V@VDDIO = 3V
LP mode						
VDDIO	Loading current	-	-	5	mA	VDDIO=3V@V _{BAT} = 4.2V

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2.3 Battery Charge

Table 2-3

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
LDOIN	Charge Input Voltage	4.5	5	5.5	V	—
V _{Charge}	Charge Voltage	4.15	4.2	4.25	V	LDOIN>4.5V
		4.30	4.35	4.40	V	
I _{Charge}	Charge Current	20		200	mA	Charge current at fast charge mode
I _{Trinkl}	Trickle Charge Current	20	45	70	mA	V _{BAT} < V _{Trinkl}

2.4 IO Input/Output Electrical Logical Characteristics

Table 2-4

IO input characteristics						
Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
V _{IL}	Low-Level Input Voltage	-0.3	—	0.3* VDDIO	V	VDDIO = 3.0V
V _{IH}	High-Level Input Voltage	0.7* VDDIO	—	VDDIO+0.3	V	VDDIO = 3.0V
IO output characteristics						
V _{OL}	Low-Level Output Voltage	—	—	0.33	V	VDDIO = 3.0V
V _{OH}	High-Level Output Voltage	2.7	—	—	V	VDDIO = 3.0V

2.5 Internal Resistor Characteristics

Table 2-5

Port	Drive Strength	Internal Pull-Up Resistor	Internal Pull-Down Resistor	Comment
PA1-PA8, PB1-PB7, PC1-PC5, PD5-PD7	drive_select[11] 64mA drive_select[10] 26.4mA drive_select[01] 8mA drive_select[00] 2.4mA	10K	10K	1. PB1 default pull up 2. USBDM&USBDP default pull down 3. Internal pull-up/pull-down resistance accuracy ±20% 4. PB8,P00 can pull-up resistance to 5V
PB8,P00	8mA	10K	10K	
USBDP	4mA	1.5K	15K	
USBDM	4mA	180K	15K	

2.6 BT Characteristics

2.6.1 Transmitter

Basic Rate

Table 2-6

Parameter		Min	Typ	Max	Unit	Test Conditions
RF Transmit Power		-	4	6	dBm	25°C, Power Supply VBAT=3.7V 2441MHz DH5
RF Power Control Range		-	20	-	dB	
20dB Bandwidth		-	950	-	KHz	
In-band spurious Emissions (BQB Test Mode RF_Tx Power=4dBm)	$F=F_0 \pm 1\text{MHz}$	-	-20	-	dBm	
	$F=F_0 \pm 2\text{MHz}$	-	-45	-	dBm	2441MHz DH5
	$F=F_0 \pm 3\text{MHz}$	-	-35	-	dBm	
	$F=F_0 \pm >3\text{MHz}$	-	-45	-	dBm	

Enhanced Data Rate

Table 2-7

Parameter		Min	Typ	Max	Unit	Test Conditions
Relative Power		-	-1	-	dB	25°C, Power Supply VBAT=3.7V 2441MHz 2DH5
$\pi/4$ DQPSK Modulation Accuracy	DEVM RMS	-	6	-	%	
	DEVM 99%	-	16	-	%	
	DEVM Peak	-	12	-	%	
In-band spurious Emissions (BQB Test Mode RF_Tx Power=4dBm)	$F=F_0 \pm 1\text{MHz}$	-	-4	-	dBm	2441MHz 2DH5
	$F=F_0 \pm 2\text{MHz}$	-	-30	-	dBm	
	$F=F_0 \pm 3\text{MHz}$	-	-30	-	dBm	
	$F=F_0 \pm >3\text{MHz}$	-	-37	-	dBm	

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2.6.2 Receiver

Basic Rate

Table 2-8

Parameter		Min	Typ	Max	Unit	Test Conditions
Sensitivity		-	-91	-	dBm	25°C,
Co-channel Interference Rejection		-	7	-	dB	
Adjacent Channel selectivity C/I	+1MHz	-	-5	-	dB	Power Supply VBAT=3.7V 2441MHz DH5
	-1MHz	-	-7	-	dB	
	+2MHz	-	-33	-	dB	
	-2MHz	-	-40	-	dB	
	+3MHz	-	-22	-	dB	
	-3MHz	-	-46	-	dB	

Enhanced Data Rate

Table 2-9

Parameter		Min	Typ	Max	Unit	Test Conditions
Sensitivity		-	-92	-	dBm	25°C,
Co-channel Interference Rejection		-	9	-	dB	
Adjacent Channel selectivity C/I	+1MHz	-	-12	-	dB	Power Supply VBAT=3.7V 2441MHz 2DH5
	-1MHz	-	-14	-	dB	
	+2MHz	-	-27	-	dB	
	-2MHz	-	-33	-	dB	
	+3MHz	-	-21	-	dB	
	-3MHz	-	-38	-	dB	

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2.6.3 BLE

1M Data Rate

Table 2-10

Parameter		Min	Typ	Max	Unit	Test Conditions
Sensitivity		-	-94	-	dBm	25°C Power Supply VBAT=3.7V 2440MHz
RF Transmit Power		-	5	-	dBm	
In-band Spurious Emission	M-N =2MHz	-	-43	-	dBm	
	M-N ≥3MHz	-	-41	-	dBm	
Modulation Characteristics	Δf1 avg	-	250	-	KHz	
	Δf2 99%	-	210	-	KHz	
	Δf1avg/Δf2avg	-	0.9	-	/	
Carrier Frequency Offset		-50	-	+50	KHz	
Frequency Drift		-25	-	+25	KHz	
Frequency Drift Rate		-5	-	+5	KHz/50us	

2M Data Rate

Table 2-11

Parameter		Min	Typ	Max	Unit	Test Conditions
Sensitivity		-	-91	-	dBm	25°C Power Supply VBAT=3.7V 2440MHz
RF Transmit Power		-	5	-	dBm	
In-band Spurious Emission	M-N =4MHz	-	-40	-	dBm	
	M-N =5MHz	-	-40	-	dBm	
	M-N ≥6MHz	-	-40	-	dBm	
Modulation Characteristics	Δf1 avg	-	500	-	KHz	
	Δf2 99%	-	430	-	KHz	
	Δf1avg/Δf2avg	-	0.9	-	/	
Carrier Frequency Offset		-50	-	+50	KHz	
Frequency Drift		-25	-	+25	KHz	
Frequency Drift Rate		-5	-	+5	KHz/50us	

Long Range

Table 2-12

Parameter	Min	Typ	Max	Unit	Test Conditions
Sensitivity LE 125K(S8)	-	-101	-	dBm	VBAT=3.7V,25°C
Sensitivity LE 500K(S2)	-	-97	-	dBm	2440MHz

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3. Package Information

3.1 QFN32(4mm*4mm)

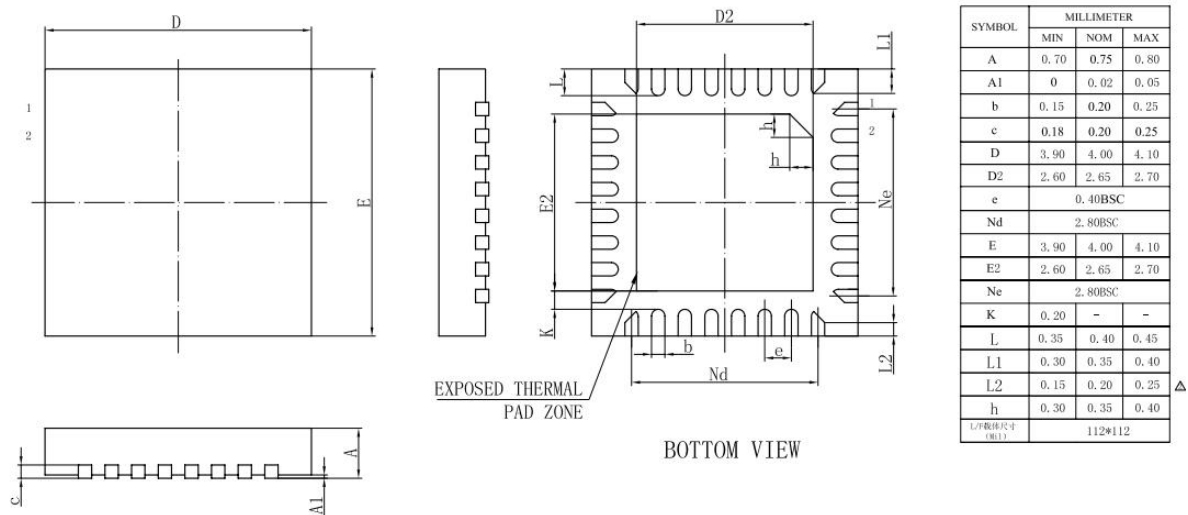
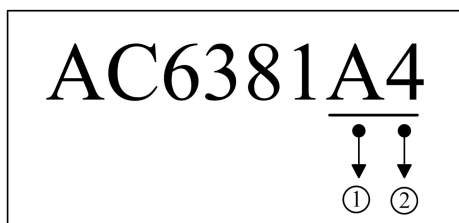


Figure 3-1 AC6381A_QFN32 Package

4. Package Type Specification



①Represents different packages

②Represents different memory sizes

4: 4Mbit Flash

5. Revision History

Date	Revision	Description
2021.05.19	V1.0	Initial Release
2022.07.19	V1.1	Update Bluetooth Feature
2024.03.06	V1.2	Update Bluetooth Feature, Add BLE Parameter
2025.01.09	V1.3	Update Bluetooth Feature, Add Pin Description Note